

# Introduction to Stochastic processes

- ▶ Stochastic process  $\{X(t), t \in T\}$  on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  is a collection of random variables defined such that for every  $t \in T$  we have  $X(t) : \Omega \rightarrow \mathcal{S}$ .
- ▶  $T$  is the parameter space (often resembles time) and  $\mathcal{S}$  is the state space.
- ▶ When  $T$  is countable, we have a discrete time process. If  $T$  is a subset of real line, we have a continuous time process.
- ▶ State space could be integers or real numbers
- ▶ State space could even be  $\mathbb{R}^n$  or  $\mathbb{Z}^n$  valued

# Examples of Stochastic Processes

- ▶ Sequence of i.i.d random variables.
- ▶ General random walk: If  $X_1, X_2, \dots$  is a sequence i.i.d of random variables, then  $S_n = \sum_{i=1}^n X_i$  is a random walk.
- ▶ Wiener process:  $\{X(t), t \geq 0\}$  is a Wiener process if
  1.  $X(0) = 0$
  2.  $\{X(t), t \geq 0\}$  has **stationary and independent increments**
  3. for every  $t > 0$ ,  $X(t)$  is normally distributed with mean 0 and variance  $t$ .
- ▶  $\{X(t), t \geq 0\}$  is a Markov process if for  $t_1 < t_2 < \dots t_n < t$  we have
$$P(X(t) \leq x | X(t_1) = x_1, \dots, X(t_n) = x_n) = P(X(t) \leq x | X(t_n) = x_n)$$
- ▶ Random walk and Wiener process are examples of Markov processes.

# Introduction to Stochastic processes

A c.t.s.p. is called an *independent increment process* if for any choice of parameters  $t_0 < t_1 < \dots < t_n$ , the  $n$  increment random variables  $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$  are independent.

The c.t.s.p. is said to have *stationary increments* if in addition  $X(t_2 + s) - X(t_1 + s)$  has the same distribution as  $X(t_2) - X(t_1)$  for all  $t_1, t_2 \in T$  and any  $s > 0$ .

# Counting process

Stochastic process  $\{N(t), t \geq 0\}$  is a counting process if it represents the total number of events upto time  $t$ .

It satisfies the following

- ▶  $N(t) \geq 0$  and is integer valued
- ▶ For  $s \leq t$ , we have  $N(s) \leq N(t)$ .  $N(t) - N(s)$  denotes the number of events in the interval  $(t, s)$
- ▶ We will now look at some examples of counting processes that have independent and stationary increments.

# Bernoulli/Binomial process

- ▶ Bernoulli( $p$ ) random variable represents an event.
- ▶ Bernoulli process is a sequence of independent r.v.'s  $\{X_i, i = 1, 2, \dots\}$  where  $X_i$  is a Bernoulli( $p$ ) random variable at time  $i$ .
- ▶ Let  $S_n$  counts the number of Bernoulli events by time  $n$ .  $S_n$  is essentially sum of  $n$  independent Bernoulli( $p$ ) variables.
- ▶  $S_n = \sum_{i=1}^n X_i$  which denotes the number of event by time  $n$  and  $P(S_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ .
- ▶  $E[S_n]$  ?  $\text{Var}(S_n)$  ?

# Bernoulli/Binomial process

- ▶  $\{S_n = \sum_{i=1}^n X_i, n = 1, 2, \dots\}$  is called as a Binomial process.
- ▶ Let  $T := \{\text{smallest } n : S_n > 0.\}$ .
- ▶  $T$  is a geometric random variable with parameter  $p$ , i.e.,  
 $P(T = n_1) = p(1 - p)^{(n_1-1)}$ .
- ▶ Memoryless property:  $P(T > m + n | T > n) = P(T > m)$ .

# Poisson process

A Poisson process with rate  $\lambda$ ,  $\lambda \geq 0$  is a counting process  $\{N(t), t \geq 0\}$  with the following properties

- ▶  $N(0) = 0$
- ▶  $N(t)$  has independent increments
- ▶ Number of events in an interval of length  $t$  is a Poisson distribution with mean  $\lambda t$ . (Hence stationary increments)
- ▶  $E[N(t+s) - N(t)] = \lambda s$

Condition 3 is difficult to verify ! Hence ...

## Poisson process – Alternative definition

A function  $f$  is said to be  $o(h)$  if  $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$ .

A Poisson process with rate  $\lambda, \lambda \geq 0$  is a counting process  $\{N(t), t \geq 0\}$  with the following properties

- ▶  $N(0) = 0$
- ▶  $N(t)$  has independent and stationary increments
- ▶  $P\{N(h) = 1\} = \lambda h + o(h)$
- ▶  $P\{N(h) \geq 2\} = o(h)$



# Poisson process

## Lemma

*Definition 1  $\implies$  Definition 2*

Proof on board.

## Lemma

*Definition 2  $\implies$  Definition 1*

Self Study: Refer Sheldon Ross, Stochastic processes, Theorem 2.1.1

## Poisson Processes Definition 3

A ctsp  $\{N(t), t \geq 0\}$  is a Poisson process with rate  $\lambda > 0$  if

- ▶  $N(0) = 0$
- ▶  $N(t)$  is a counting process with stationary and independent increments
- ▶  $X_i$ , the time interval between  $i - 1$ th and  $i$ th event is exponentially distributed with parameter  $\lambda$ .